

Real-Time Monitoring and Prediction of Fruit Ripening

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Abstract

In the modern food supply chain, effective monitoring of fruit ripening processes is crucial for reducing waste, ensuring quality, and optimizing inventory management. Today, warehouses, transportation systems, and supermarkets often lack real-time insights into the microclimatic conditions surrounding fresh produce. This gap can lead to premature ripening or spoilage of fruits, improper storage management, and inefficient resource allocation across the distribution network. Real-time monitoring and predictive capabilities, as explored in this project, ultimately enhance profitability and sustainability.

1 Introduction

The quality and shelf life of fresh produce, particularly fruits, are critical factors in the global food supply chain (1). Fruit ripening is a complex physiological process influenced by multiple factors, such as temperature, humidity, and the presence of various gases, including volatile organic compounds (VOCs) (2; 3). The delicate balance of these elements not only affects the fruit’s appearance and taste but also plays a key role in its shelf life and nutritional value (4). Poor control over the ripening process can lead to early spoilage, resulting in significant economic losses and food waste (5).

Recent consumer trends emphasizing freshness and minimal use of chemical preservatives have heightened the need for advanced monitoring of products from harvest to sale (6). The efficiency of storage, transportation, and display methods in supermarkets depends largely on maintaining optimal environmental conditions (7). This need is amplified by the growing complexity of supply chains, where fruits often pass through multiple stages before reaching consumers. Sudden changes in storage conditions can accelerate ripening or degrade quality, challenging traditional monitoring approaches (8).

Furthermore, the competitive global market demands improved decision-making for managing perishable goods. With fluctuations in environmental parameters and varying conditions in storage facilities or transportation, stakeholders require accurate, timely data to prevent quality decline (9; 10). Such data can form the basis for better management practices and strategic interventions to maintain ripening balance, ensuring products reach consumers in optimal condition (11).

In the modern food supply chain, effective monitoring of fruit ripening is thus essential for reducing waste, ensuring quality, and optimizing inventory management (12). However, warehouses, transportation systems, and supermarkets often lack real-time insights into microclimatic conditions around fresh produce (13; 14; 15). This gap can result in premature ripening or spoilage, improper storage management, and inefficient resource allocation across distribution networks. In summary, the current state of post-harvest management and supply chain logistics reveals a significant shortfall in monitoring dynamic processes that influence fruit ripening (1). With rising global standards for quality and sustainability, the primary challenge is bridging the divide between understanding critical environmental factors and managing them effectively to preserve perishable goods throughout the supply chain (6).

This project addresses these challenges by introducing an Internet of Things (IoT)-based solution for real-time monitoring and analysis of key environmental variables that affect fruit ripening (13; 8). The system employs a microcontroller connected to gas sensors, as well as temperature and humidity sensors, to collect live data on VOCs, carbon dioxide, hydrogen, ethanol, temperature, and humidity (16; 5). The collected data is transmitted to a central server, stored in a database, cleaned, and pre-processed for subsequent analysis, while being mindful of potential data leakage issues in mining processes (17).

By applying regression analysis techniques and long short-term memory models for prediction, the project seeks to uncover correlations between gas emissions and ripening stages (18). The predictive insights generated enable stakeholders to anticipate changes in fruit qual-



Figure 1: Metric for Banana Ripening Stages

ity, facilitating proactive measures in warehouse management, transportation logistics, or supermarket environments (10). This approach improves storage practices and prevents unnecessary waste, ultimately enhancing profitability and sustainability.

Overall, this project’s contribution lies in providing a comprehensive, real-time monitoring system that integrates advanced sensing technology with data mining and machine learning methods (12). This unified framework not only improves tracking of environmental conditions impacting fruit ripening but also equips decision-makers with predictive analytics for timely interventions and greater supply chain efficiency (7).

2 Evaluation Criteria

The evaluation criteria for this project are based on the correlation between data collected from sensors and the fruit ripening stages observed by humans. Specifically, the system uses optical sensors to detect color changes in bananas, which serve as a key visual indicator of ripening, as shown in Figure 1. At the same time, the Brix value (the percentage by weight of solids in a solution) is measured using a refractometer to assess sugar levels. By comparing these objective measurements with standardized human assessments of fruit ripeness, reliable thresholds can be set for classifying ripening stages. The system’s accuracy and predictive performance will be assessed based on how well its outputs match the observed color changes and Brix readings, confirming its reliability and precision in defining and monitoring the fruit ripening process.

3 Dataset

The datasets used in this project are collected through a network of sensors placed in simulated environments related to fruit storage and transportation. This data gathering process involved more than 15 experiments conducted over the last year, tracking bananas from their raw state until spoilage. Several factors influence the VOC values, including temperature and humidity, with different seasons playing a key role, as shown in Figure 2—for instance, VOC levels tend to be higher and

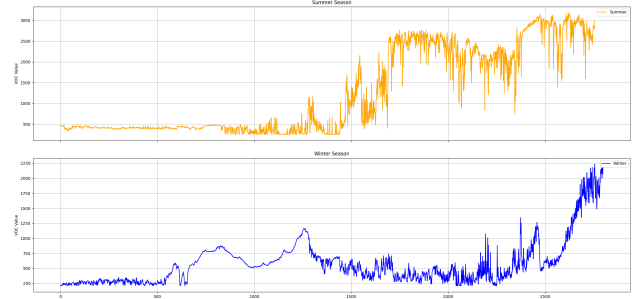


Figure 2: Tracking TVOC Values Across Summer and Winter Seasons

exhibit more noise in hotter seasons. The main data sources include measurements from the SGP30 sensor, which monitors volatile organic compounds and gases such as carbon dioxide, hydrogen, and ethanol, as well as two temperature and humidity sensors (HTU21 and SHT20) that record environmental temperature and humidity, as shown in Figure 3. Each record in the dataset includes a timestamp, precise sensor readings, and identifiers for the location or device involved.

In addition, the dataset includes validation measurements to assess the system’s predictive abilities in detecting fruit ripening trends. For instance, color sensor data is used to detect changes in banana skin color, a widely accepted visual indicator of ripening, and refractometer readings are collected to obtain Brix values. These extra data points serve as real benchmarks for evaluating the accuracy of sensor readings and refining predictive models.

Raw sensor data goes through preprocessing steps, such as cleaning to remove noise, handling missing values, and synchronizing data across different sensors. These preprocessing steps are vital to ensure the dataset’s integrity and reliability, so it can accurately reflect the dynamic environmental conditions that affect fruit ripening. Through this careful, multi-dimensional data collection and follow-up processing, the project aims to build a strong foundation for understanding correlations between environmental parameters and fruit ripening, paving the way for effective predictive analysis in real-world applications.

4 System Design

4.1 Data Collection and Storage in the System

The system’s data storage process begins at the sensor level, where environmental parameters critical to fruit ripening are captured in real-time using specialized sensors integrated with an ESP8266 NodeMCU microcontroller. Sensors such as the SGP30 detect volatile organic compounds (VOCs) and gases including carbon dioxide (CO_2 in ppm), hydrogen (in ppm), and ethanol (in ppm),



Figure 3: Picture of the Data Gathering Board

which serve as indicators of ripening-related emissions, while HTU21 and SHT20 sensors measure temperature ($^{\circ}\text{C}$) and humidity (RH) to monitor ambient conditions. These sensors connect to the microcontroller via the I²C protocol, allowing for efficient, low-power data transfer. The microcontroller collects readings at configurable intervals (e.g., every few seconds to minutes), performs initial preprocessing such as hourly auto-calibration for VOC sensors (using ambient temperature and humidity data to adjust for environmental drift and ensure accuracy) and threshold filtering to discard outliers (e.g., VOC levels exceeding 5000 ppb or temperatures above 50°C , based on empirical thresholds from sensor documentation). Validated data is then buffered in the microcontroller’s onboard memory (e.g., flash storage) before being packaged into compact payloads and transmitted wirelessly via HTTP or MQTT protocols to a central server for long-term storage in a database like SQLite, ensuring a historical record with timestamps, sensor IDs, and metadata for subsequent analysis.

4.2 Preprocessing and Preparation for Prediction

Following data collection and initial storage, preprocessing is applied to refine the raw sensor data into a reliable dataset suitable for predictive modeling. On the microcontroller, basic filtering removes noise and implausible readings, while server-side processing employs advanced techniques such as Z-score analysis, median absolute deviation (MAD), and clustering-based outlier detection to identify and eliminate anomalies caused by sensor faults or environmental interference. Data is then aggregated

(e.g., into hourly averages) and synchronized across sensors to create a cohesive time-series stream. This preprocessed data forms the foundation for the ripening prediction module, where it is enriched with labeled data to train and evaluate models. Labeled data includes ground-truth measurements like Brix values (a refractometric index quantifying soluble sugar content in fruit, typically ranging from 8-12% for ripe stages, obtained via manual sampling or dedicated Brix meters) and color sensor readings (e.g., RGB or HSV values from sensors like TCS3200, capturing color transitions from unripe green to ripe red/yellow hues). These labels are aligned with timestamped sensor data to create supervised datasets, enabling models to learn correlations between environmental inputs (e.g., gas emissions, temperature) and ripening outcomes.

4.3 Ripening Prediction Using Regression Models

In the ripening prediction phase, simple linear regression models are employed to establish straightforward relationships between preprocessed sensor data and labeled ripening indicators. The regression model treats environmental features (e.g., CO_2 levels, humidity, and VOC concentrations) as independent variables, with labeled Brix values or color indices as the dependent variable to predict. For instance, a linear equation might be fitted as $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$, where $\hat{y}$ represents predicted Brix (e.g., sugar maturity level) and x_i are sensor inputs; training minimizes mean squared error using historical labeled data. This approach generates predictions like estimated time-to-ripeness (e.g., 3-5 days based on Brix progression from 6% to 10%), visualized in charts such as scatter plots showing actual vs. predicted Brix values over time or line graphs illustrating linear trends in color shifts correlated with gas emissions. Regression is computationally efficient for real-time inference on the server, providing interpretable results that highlight key factors like temperature’s impact on ripening speed, with periodic retraining using new labeled samples to adapt to different fruit types.

4.4 Ripening Prediction Using LSTM Models

For more complex temporal patterns, long short-term memory (LSTM) neural networks are integrated to capture sequential dependencies in the time-series data, outperforming regression in scenarios with non-linear or lagged effects. The LSTM model processes sequences of preprocessed sensor data (e.g., sliding windows of hourly aggregates) as input features, with labeled Brix and color values serving as targets for multi-step forecasting. During training, the network’s architecture—comprising input layers, LSTM cells with forget/input/output gates to handle long-term dependencies, and dense output

layers—learns to predict future ripening states by minimizing loss functions like mean absolute error on labeled datasets (e.g., sequences where Brix increases from 7% to 11% alongside rising ethanol levels). Predictions are output as time-series charts, such as multi-line graphs depicting actual and forecasted Brix curves or color index trajectories over days, enabling estimates like “fruit will reach optimal ripeness in 48 hours” with confidence intervals. The model is pre-trained offline on historical labeled data and deployed on the server for real-time inference, with fine-tuning capabilities using accumulated new data to improve accuracy in varying environmental conditions, making it ideal for dynamic supply chain monitoring.

5 Conclusion

This project presents a comprehensive IoT-based solution for monitoring and analyzing factors influencing fruit ripening, with a special focus on gases and environmental conditions. By leveraging sensor modules for volatile organic compounds and temperature/humidity sensors connected to an ESP8266 NodeMCU microcontroller, the system collects real-time data reflecting critical ripening indicators such as gas concentrations, temperature, and humidity. Beyond data collection, the microcontroller employs initial threshold checks to filter improbable readings and refines valid data before transmitting it to the network.

The predictive modeling component consists of simple regression approaches or an advanced long short-term memory (LSTM) network. These models are trained using historical sensor readings alongside ground-truth data (color sensor and Brix measurements) to forecast fruit ripening trends. In real-time mode, incoming data from the server is fed into the pre-trained model, and the resulting predictions can lead to immediate decision-making, such as adjusting storage conditions or planning inventory turnover to prevent fruit spoilage.

By integrating continuous data collection, robust data management, and predictive analytics into a unified platform, this project addresses a critical gap in the fresh produce supply chain. The system enables stakeholders to observe environmental conditions at various points and perform timely interventions based on data. The detailed logging and predictive capabilities of the system facilitate rapid responses to potential quality declines in fruit, ultimately leading to reduced food waste and increased operational efficiency.

6 Future Work

Beyond laboratory or simulated environments, the next phase involves deploying the system in fully operational warehouses. This will help evaluate sensor performance and predictive model accuracy under varying storage

practices, high inventory volumes, and real-world environmental fluctuations.

Experiments in different temperature and humidity profiles, particularly in colder and warmer regions, will be conducted. This will assess the system’s resilience and enable fine-tuning of parameters (such as sensor sampling frequency or data preprocessing thresholds) to meet diverse logistical needs.

While bananas were selected as the initial case, future experiments will extend to fruits like tomatoes and pineapples, as these also produce volatile organic compound gases that can indicate ripening stages. Due to the unique physiological processes of each fruit, gas profiles and environmental requirements may differ significantly.

The system’s prediction component can be customized or retrained to incorporate fruit-specific variables. For example, tomatoes differ from bananas in ethylene production and response, requiring adjusted thresholds and predictive models. Testing these unique signatures will enhance the platform’s overall adaptability.

Investigating fruit ripening processes over longer time frames, such as several weeks, will provide deeper insights into how changes accumulate during storage periods. This could reveal subtle sensor drifts or patterns not observable in short-term experiments.

The microcontroller could adjust sampling frequency based on observed conditions. For instance, in stable environments, the system might reduce sampling to conserve energy, but increase it during periods of high variability.

In addition to linear regression and LSTM networks, future work could incorporate ensemble techniques or transformer-based models to better understand complex relationships between gases and ripening trends.

Integrating advanced algorithms (such as autoencoders or density-based methods) could improve outlier detection, especially in high-noise sensor environments or noisy settings.

By linking ripening trend predictions to inventory management and transportation scheduling, warehouses and distributors can optimize delivery routes and storage durations based on predicted fruit quality.

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